

Diagnostic Framework: Three Levels of Value-Sensitivity Testing

Does normative-frame sensitivity affect LLM-assisted policy simulation?

H1: Model-Level

Composite Orientation-Effect

Aggregate PVC index →
simulated burden effect

N = 15 model endpoints

$\beta_1 = 1.51$, 95% CI [-2.41, 5.43]

$R^2 = 0.044$

Null: no composite pathway

H2: Narrative-Level

Frame Association

Expressed rationale frames →
simulated burden effect

N = 15 × 27 profiles

Access barriers: $b = 8.83$, $p < 0.001$ (↑)

Collective resp.: $b = -6.20$, $p < 0.001$ (↓)

Coercive backlash: $b = 6.02$, $p < 0.001$ (↑)

**Frame decomposition confirms
narrative mechanism**

H3: Within-Model

Frame Manipulation

Assigned governance-value frame →
simulated burden effect

N = 4 × 27 × 4

Autonomy: $\Delta = +2.65$, $p < 0.001$ (↑)

Equity/access: $\Delta = +6.27$, $p < 0.001$ (↑)

Collective obligation: $\Delta = -10.54$, $p < 0.001$ (↓)

**Causal manipulation;
adj. $R^2 = 0.66$**

independent diagnostic tests

Three independent diagnostic tests — each informative regardless of the others

PVC null (H_1) → narrative frames matter (H_2) → causal manipulation confirms (H_3)